Communication-Aware Adaptive Weights for Consensus-Oriented Distributed KLA-Based LMB Fusion

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ABSTRACT

Distributed multi-sensor multi-object tracking over peer-to-peer networks requires conservative fusion under unknown cross-correlations, but fixed consensus weights do not reflect time-varying posterior quality or realized packet delivery. This paper proposes a communication-aware adaptive weighting rule for consensus-oriented geometric-average (GA) realization of Kullback-Leibler average (KLA) fusion of labeled multi-Bernoulli (LMB) posteriors. Each node scores available neighbors by posterior concentration, delivered-versus-dropped communication quality, and Bernoulli existence decisiveness. The score is then split into spatial and existence branches: covariance and link-quality cues drive Gaussian spatial fusion, existence confidence drives Bernoulli support fusion, and a Fisher information distance and Fisher information accumulation (FID-FIA) cue is optionally applied only to the existence branch. This yields two operating modes. Balanced mode uses the low-overhead branch-decoupled backbone, whereas Cardinality-critical mode activates the additional existence-only FID-FIA refinement when target-number agreement is the main risk. In an eight-sensor dual-formation scenario with finite fields of view and tiered heterogeneous packet loss, the proposed modes reduce network-level disagreement relative to fixed and dynamic-weight baselines while retaining conventional local tracking accuracy as a safeguard. The method therefore treats adaptive distributed GA-LMB fusion as an explicit spatial-consensus, cardinality-consensus, and computational-cost tradeoff.

1. Introduction

Distributed multi-sensor multi-object tracking is central to surveillance, autonomous systems, and cooperative perception. The difficult case considered here is not centralized multi-view fusion with a reliable communication backbone, but peer-to-peer tracking in which nodes have partial field-of-view (FOV) coverage, time-varying posterior quality, and communication affected by bandwidth limits, packet drops, or temporary outages [1]. Random finite set (RFS), finite-set statistics (FISST), and labeled-RFS filters provide a Bayesian representation for target number, kinematic state, and identity [2–6]. The remaining fusion question is how neighboring labeled posteriors should be combined when FOV geometry, posterior uncertainty, and packet delivery all vary across the peer-to-peer network.

Within this family, generalized labeled multi-Bernoulli (GLMB) and labeled multi-Bernoulli (LMB) filters balance multi-object modeling fidelity and computational tractability [5–9]. For distributed fusion, the geometric-average (GA) realization of Kullback-Leibler average (KLA), also known as generalized covariance intersection (GCI), gives a conservative logarithmic-opinion-pool rule when unknown cross-correlations make exact Bayesian fusion unsafe [10–15]. This makes GA/KLA-based LMB fusion a natural backbone for peer-to-peer multi-object tracking; the unresolved design question is how to allocate its weights online.

Fixed Metropolis, uniform, or degree-based consensus weights [16] are easy to deploy, but they cannot distinguish precise from ambiguous posteriors, or delivered neighbors from nominally connected but unreliable ones. A single adaptive reliability score is also too coarse for an LMB posterior: a Bernoulli component may be spatially sharp but existence-uncertain, or existence-decisive but spatially diffuse. The missing piece is a factorized and branch-aware weight allocation rule for communication-constrained peer-to-peer GA-LMB fusion.

Motivated by this gap, this paper proposes a communication-aware adaptive fusion-weight allocation family for distributed GA-LMB fusion; Figure 1 summarizes the resulting peer-to-peer update. The design builds a shared

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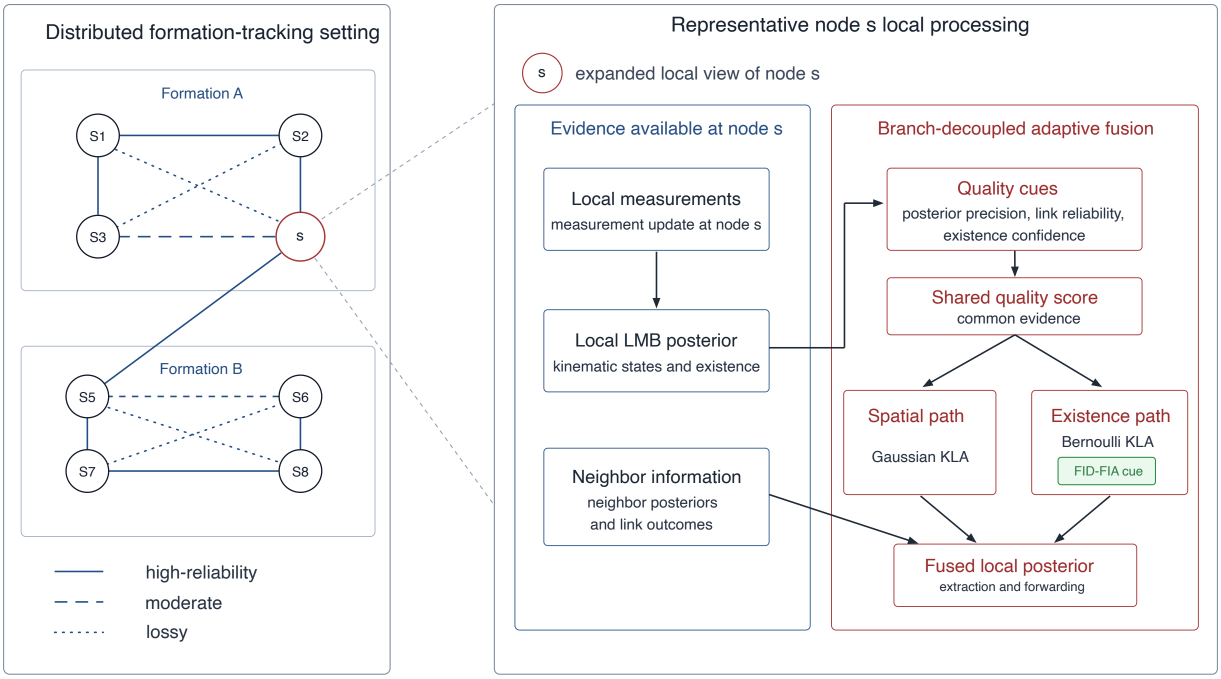

Figure 1: System overview of the distributed eight-sensor tracking setting, including local measurement update, neighborhood communication, branch-decoupled adaptive GA-LMB fusion, and an FID-FIA cue applied only on the existence-weight path.

backbone from covariance quality, realized link quality, and existence confidence, then separates the spatial and existence branches of the LMB fusion rule. A weak structure-aware correction regularizes spatial fusion without letting topology dominate the posterior-quality terms, while a Fisher information distance and Fisher information accumulation (FID-FIA) refinement imports the information-geometric cue of Cao and Zhao [17] only into the Bernoulli existence branch. This leads to two operating modes rather than a single claimed optimum: a Balanced mode for spatial/position-sensitive deployment and a Cardinality-critical mode for target-number agreement.

The evaluation uses an eight-sensor dual-formation tracking scenario under tiered heterogeneous packet loss as a controlled stress test, not as an independent novelty claim. It exposes geometry/FOV-induced posterior-quality variation and realized communication heterogeneity. Network-level disagreement metrics are the primary outcome because the method is designed to reduce disagreement in the fused multi-sensor picture; truth-referenced local metrics and filter/fusion runtime serve as safeguards against hidden accuracy or deployment-cost regressions. The results show the intended tradeoff: communication-aware adaptive weighting improves consensus over fixed and dynamic-weight baselines, the covariance/link/existence backbone gives most of the low-overhead gain, and the existence-only FID-FIA refinement is most useful when cardinality agreement is the dominant risk.

The main contributions of this paper are as follows.

1. We propose a communication-aware adaptive fusion-weight allocation family for distributed GA/KLA-based LMB fusion under communication constraints.
2. We introduce a factorized quality model with posterior covariance, realized link quality, and existence confidence, together with branch-specific structure-aware and FID-FIA refinements.
3. We give an explicit per-node algorithmic flow and design rationale so that the coefficients are presented as bounded reliability cues rather than disconnected tuning factors.
4. We show, with tiered-loss, ideal-communication, communication-sensitivity, and runtime evidence, that the method improves network-level consensus while exposing low-overhead Balanced and higher-cost Cardinality-critical operating choices.

2. Related Work

The most relevant prior work falls into three groups: labeled-RFS filtering, conservative distributed density fusion, and adaptive or weighted RFS/LMB fusion. The first group provides the statistical state representation. Mahler's RFS/FISST formulation and probability hypothesis density (PHD) recursion establish the set-valued Bayesian foundation [2–4], while GLMB and LMB filters add labeled multi-object posteriors suitable for track identity and tractable implementation [5–9]. The present paper does not modify this filtering family; it uses LMB posteriors as the objects to be fused.

The second group concerns conservative fusion when cross-correlations are unknown. Covariance intersection, Kullback-Leibler averaging, geometric-average fusion, and inverse-covariance-intersection variants provide density-combination rules that avoid overconfident fusion under unknown dependence [10–13, 18–20]. Related logarithmic-opinion-pool and exponential-mixture formulations connect the same idea to distributed Bayesian fusion and consensus [14, 15]. In the labeled-RFS setting, GCI/KLA-style fusion has been extended to multi-Bernoulli and labeled-RFS densities, including work on label mismatch and peer-to-peer resilience [21–23]. These studies justify the GA/KLA backbone used here, but they leave open how the weights should respond to time-varying posterior quality and realized communication quality.

The third group is most directly related to adaptive weighting. Centralized multiple-view LMB fusion shows that information-aware weights can help when sensors have limited or different fields of view [24, 25]. Klupacs et al. take a complementary-fusion view, fusing label-associated Bernoulli densities and tuning sensor contribution by object detectability [26]. Consensus-based distributed LMB tracking has also explored automatic track-level weight evolution for different fields of view [27]. Recent peer-to-peer RFS work moves still closer to the present setting by factorizing local RFS densities into subdensities and assigning local weighting coefficients before GA or arithmetic-average (AA) fusion under consensus or flooding protocols [28]. The closest information-geometric GA/KLA precedent is Cao and Zhao's distributed heterogeneous labeled-RFS method, which uses Fisher information distance and Fisher information accumulation (FID-FIA) to assign adaptive scalar fusion weights under heterogeneous sensing architectures [17]. In parallel, AA density fusion and related minimum-information-loss or cardinality-consistent formulations clarify how arithmetic-average and geometric-average fusion emphasize different objectives and operating behavior [29–32]. A recent TechRxiv preprint in the AA-density-fusion series further studies Fisher-information-proportional weights for time-aligned asynchronous multi-rate filters and reports their use in both AA and GA fusion [33].

Communication constraints provide the deployment motivation for the weighting problem. Bandwidth limits, delays, asynchronous updates, and packet losses are central design factors in distributed fusion over sensor networks [1]. The present work focuses on the packet-delivery side of this problem: realized communication quality enters directly into the fusion weights, rather than appearing only as a nominal graph topology or an external simulation condition.

Against this background, the contribution of this paper is deliberately narrower than a broad new fusion framework. We stay within peer-to-peer GA/KLA-based LMB fusion and focus on the online weight-allocation mechanism. Relative to fixed-weight GA/KLA fusion, the weights react to posterior concentration and realized packet delivery. Relative to adaptive scalar-weight formulations, the rule separates the spatial and existence branches of the LMB posterior and confines the optional FID-FIA cue to the existence path.

3. Problem Formulation

Consider a distributed sensor network with sensor index set $\mathcal{V} = \{1, \dots, S\}$, where S is the number of sensors, operating over discrete time $k \in \{1, \dots, T\}$. The multi-object state at time k is modeled as a labeled random finite set X_k, and each sensor $s \in \mathcal{V}$ acquires a local measurement set $Z_k^{(s)}$. Following the standard RFS/FISST and labeled-RFS formulation, each node maintains an LMB approximation to the local multi-object posterior, whose Bernoulli components jointly encode target existence, kinematic state, and identity [2, 4–7]. The objective of the distributed fusion layer is to combine neighboring local posteriors into a consensus-quality posterior without assuming that cross-correlations among nodes are known.

3.1. Local LMB posterior model

Let $\mathcal{N}_s \subseteq \mathcal{V}$ denote the communication neighborhood of sensor s, including the node itself. In the setting considered here, $\mathcal{N}_s$ is induced by a communication graph derived from sensor geometry and communication range, and each node performs local fusion only over the posteriors available inside this neighborhood. After local prediction, measurement update, and data association, node $j \in \mathcal{N}_s$ produces a measurement-updated LMB posterior. We use $\Theta_k^{(j)}$ for its finite parameter set,

$$\Theta_k^{(j)} = \left\{ \left(r_{k,i}^{(j)}, p_{k,i}^{(j)}(\cdot, \ell_i) \right) \right\}_{i=1}^{M_k}, \tag{1}$$

where $\mathcal{L}_k = \{\ell_1, \dots, \ell_{M_k}\}$ is the aligned label set at time k, M_k is the number of aligned Bernoulli labels, $r_{k,i}^{(j)} \in [0, 1]$ is the Bernoulli existence probability, and $p_{k,i}^{(j)}$ is the associated spatial density for label ℓ_i. The corresponding LMB density is not this parameter set itself; it is the standard labeled-RFS density induced by the parameter set,

$$\pi_k^{(j)}(X) = \Delta(X)\, w_k^{(j)}(\mathcal{L}(X)) \prod_{(x,\ell_i)\in X} p_{k,i}^{(j)}(x, \ell_i), \tag{2}$$

with

$$w_k^{(j)}(L) = \prod_{\ell_i \in L} r_{k,i}^{(j)} \prod_{\ell_i \in \mathcal{L}_k \setminus L} \left(1 - r_{k,i}^{(j)}\right), \qquad L \subseteq \mathcal{L}_k. \tag{3}$$

Here $\Delta(X)$ is the distinct-label indicator and $\mathcal{L}(X)$ is the set of labels present in X. The compact parameter-set notation $\Theta_k^{(j)}$ is used below whenever the implementation operates on the Bernoulli components rather than on the full finite-set density.

3.2. Communication-constrained delivery

Before local measurements are used by a neighboring fusion node, they pass through a communication layer with three possible effects:

1. a global or per-step bandwidth budget that limits how many measurements can be delivered,
2. a stochastic packet-drop model on each sensor link,
3. optional node-level outages.

Accordingly, the posterior that reaches a neighbor is influenced not only by the underlying sensing quality, but also by whether measurements were actually delivered. Let $m_k^{(j)} \in \{0, 1\}$ denote the availability indicator of node j at time k, and let $\rho_k^{(j)} \in [0, 1]$ denote the realized link-quality statistic derived from delivered-versus-dropped packets.

3.3. Distributed KLA fusion objective

Because the cross-correlation structure among neighboring LMB posteriors is unknown, exact Bayesian fusion is generally unavailable. We therefore adopt the geometric-average realization of Kullback-Leibler averaging, also known as generalized covariance intersection, as the main conservative fusion rule. At node s, the target fused density over the neighborhood $\mathcal{N}_s$ is

$$\bar{\pi}_k^{(s)}(X) \propto \prod_{j\in\mathcal{N}_s} \left(\pi_k^{(j)}(X)\right)^{\omega_{k,s}^{(j)}}, \qquad \sum_{j\in\mathcal{N}_s} \omega_{k,s}^{(j)} = 1, \qquad \omega_{k,s}^{(j)} \geq 0, \tag{4}$$

where $\omega_{k,s}^{(j)}$ is the fusion weight assigned by node s to neighbor j at time k [12, 13]. Unless stated otherwise, the shorthand KLA/GA in this paper refers to this geometric-average/logarithmic-pool solution of the Kullback-Leibler averaging problem. Appendix AA experiments are treated separately as arithmetic-average density fusion, which is connected to the reverse-KL/minimum-information-loss route rather than to the logarithmic pooling operator in (4) [29–31]. With fixed-weight fusion, $\omega_{k,s}^{(j)}$ is often chosen as a uniform or Metropolis-type consensus weight [16]. The central problem studied here is how to choose these weights adaptively so that they reflect both posterior quality and communication quality.

In the GA-LMB fusion rule considered here, the spatial and existence parts of each Bernoulli component are fused separately after moment matching. This separation makes it natural to ask whether the same scalar weight should govern both spatial fusion and existence fusion.

3.4. Adaptive weight-allocation target

The adaptive layer must output active-neighbor weight vectors on the simplex. In the fixed-weight case, the same vector $\omega_{k,s}^{(j)}$ can be used for the whole posterior. In the branch-decoupled case studied here, the output is a spatial vector $\omega_{k,s}^{x,(j)}$ and an existence vector $\omega_{k,s}^{r,(j)}$. Both are normalized over the available neighborhood, but they are allowed to respond to different reliability cues because the Gaussian spatial update and the Bernoulli existence update consume the weights in different algebraic blocks. Section 4 gives the concrete score definitions and the per-node algorithmic flow; this section only states the fusion target and the quantities that the adaptive layer must produce.

3.5. Evaluation target

The aim of the above formulation is not only to improve each node's local tracking output, but also to improve agreement among nodes. This distinction is central in the present problem. Under unknown cross-correlations and communication losses, distributed fusion is useful only if neighboring nodes move toward a shared multi-sensor picture rather than drifting toward mutually inconsistent local beliefs.

Accordingly, the evaluation uses two complementary metric layers:

- primary network-level disagreement metrics, which quantify consensus error among the fused outputs of different sensors,
- secondary truth-referenced local tracking metrics, which verify that consensus-error reductions are not obtained by materially degrading conventional tracking accuracy.

Let $\hat{X}_k^{(s)}$ denote the extracted multi-object estimate at sensor s and time k, and let $\hat{n}_k^{(s)} = |\hat{X}_k^{(s)}|$ denote its estimated cardinality. The paper defines the following network-level disagreement metrics:

$$D_{\mathrm{OSPA}}(k) = \frac{2}{S(S-1)} \sum_{1\le a<b\le S} d_{\mathrm{OSPA}}\left(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}\right), \tag{5}$$

$$D_{\mathrm{loc}}(k) = \frac{2}{S(S-1)} \sum_{1\le a<b\le S} d_{\mathrm{loc}}\left(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}\right), \tag{6}$$

$$D_{\mathrm{card}}(k) = \frac{1}{S} \sum_{s=1}^{S} \left|\hat{n}_k^{(s)} - \mathrm{median}\left(\hat{n}_k^{(1)}, \dots, \hat{n}_k^{(S)}\right)\right|, \tag{7}$$

where d_{OSPA} is the pairwise optimal subpattern assignment (OSPA) distance between two estimated finite sets and d_{loc} is the matched localization disagreement obtained after Hungarian matching on the common extracted objects [34–36]. In the paper these are reported as OSPA consensus error, matched localization disagreement, and cardinality dispersion, respectively.

These three quantities are task-driven network-level disagreement measures rather than standard field-wide tracking benchmarks. Their building blocks are standard or conventional: OSPA and related generalized OSPA (GOSPA)-style distances are mainstream finite-set tracking metrics, Hungarian matching is the standard assignment primitive behind many tracking scores, and pairwise dispersion is the usual way to quantify disagreement in consensus problems [16, 34, 35, 37]. What is specific to this paper is the aggregation target: instead of comparing each sensor only with ground truth, the metrics compare the post-fusion outputs of different nodes. They are introduced because the present problem is distributed fusion under heterogeneous communication, where the main question is whether neighboring nodes move toward a common fused picture. For that reason, the paper does not use them as replacements for conventional truth-referenced tracking metrics, and it pairs the network-level disagreement metrics with the secondary local metrics listed below.

The secondary local metrics remain the standard truth-referenced tracking quantities:

- local expected OSPA (E-OSPA),
- local root mean square error (RMSE),
- local cardinality error.

The evaluation logic is therefore asymmetric by design: the method is primarily successful if it reduces network-level disagreement, while the local metrics are used as a safeguard to show that this gain is not purchased by a clear loss in conventional tracking quality.

4. Communication-Aware Adaptive KLA Fusion

This section describes the adaptive fusion rule used in the distributed GA-LMB filter. The starting point is the conservative geometric-average realization of Kullback-Leibler averaging for neighboring LMB posteriors under unknown cross-correlations [12, 13]. The main methodological question is not how to redesign the underlying labeled-RFS filter family, but how to allocate fusion weights so that they reflect posterior quality and realized communication quality in a communication-constrained sensor network. The scores below should be read as bounded reliability cues consumed by the GA-LMB merger, not as independent optimality claims for each coefficient. A variational and information-geometric interpretation of the branch-decoupled update is given in Appendix C.

4.1. Baseline GA-LMB fusion

At node s, let $\mathcal{N}_s$ denote the local communication neighborhood and let $\pi_k^{(j)}$ be the measurement-updated LMB posterior density provided by node $j \in \mathcal{N}_s$. The conservative fusion target is the weighted geometric average in (4). The remaining task is to specify how the LMB parameter sets are moment-projected and how the spatial and existence weights are consumed by the merger.

After moment projection, the Gaussian approximation of the ith Bernoulli spatial density at node j is

$$p_{k,i}^{(j)}(x) \approx \mathcal{N}\left(x; v_{k,i}^{(j)}, T_{k,i}^{(j)}\right). \tag{8}$$

Using spatial fusion weights $\omega_{k,s}^{x,(j)}$, the fused spatial density is obtained in canonical form:

$$K_{k,i}^{(s)} = \sum_{j\in\mathcal{N}_s} \omega_{k,s}^{x,(j)} \left(T_{k,i}^{(j)}\right)^{-1}, \qquad h_{k,i}^{(s)} = \sum_{j\in\mathcal{N}_s} \omega_{k,s}^{x,(j)} \left(T_{k,i}^{(j)}\right)^{-1} v_{k,i}^{(j)}, \tag{9}$$

$$\Sigma_{k,i}^{(s)} = \left(K_{k,i}^{(s)}\right)^{-1}, \qquad \mu_{k,i}^{(s)} = \Sigma_{k,i}^{(s)} h_{k,i}^{(s)}. \tag{10}$$

Using existence fusion weights $\omega_{k,s}^{r,(j)}$, the Bernoulli existence probability is fused by the corresponding weighted geometric rule:

$$r_{k,i}^{(s)} = \frac{\eta_{k,i}^{(s)} \prod_{j\in\mathcal{N}_s} \left(r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}}}{\eta_{k,i}^{(s)} \prod_{j\in\mathcal{N}_s} \left(r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}} + \prod_{j\in\mathcal{N}_s} \left(1 - r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}}}, \tag{11}$$

where $\eta_{k,i}^{(s)}$ is the Gaussian normalization term induced by the fused spatial density.

4.2. Factorized adaptive weight backbone

The adaptive design begins with a shared quality backbone. For node s and neighbor $j \in \mathcal{N}_s$, define the raw score

$$\tilde{\omega}_{k,s}^{(j)} = m_k^{(j)} \cdot q_{\mathrm{cov},k}^{(j)} \cdot q_{\mathrm{link},k}^{(j)} \cdot q_{\mathrm{exist},k}^{(j)}, \tag{12}$$

where $m_k^{(j)} \in \{0, 1\}$ is the availability mask, $q_{\mathrm{cov},k}^{(j)}$ is a covariance-quality score, $q_{\mathrm{link},k}^{(j)}$ is a realized link-quality score, and $q_{\mathrm{exist},k}^{(j)}$ is an existence-confidence score. The normalized weight is

$$\bar{\omega}_{k,s}^{(j)} = \frac{\tilde{\omega}_{k,s}^{(j)}}{\sum_{u\in\mathcal{N}_s} \tilde{\omega}_{k,s}^{(u)}}. \tag{13}$$

This factorization focuses on the three factors that are most directly aligned with the design objective: posterior concentration through covariance, communication reliability through delivered-versus-dropped packets, and existence decisiveness through Bernoulli existence probabilities. Figure 2 shows how this shared backbone feeds the decoupled spatial and existence-weight paths used by the two operating modes.

Remark 1 (backbone rationale). The product form is a compact way to combine independent reliability cues on the simplex. In log form, it adds the utilities $\log q_{\text{cov}}$, $\log q_{\text{link}}$, and $\log q_{\text{exist}}$, so a poor value in any cue cannot be completely hidden by a large value in another cue. The availability mask $m_k^{(j)}$ is a hard gate for an unavailable neighbor, whereas the positive floors used in some cue mappings are soft safeguards for uncertain but still usable posteriors. If all active raw scores vanish, the implementation falls back to the uniform distribution over active neighbors before applying any smoothing or final floor.

4.3. Covariance and link-quality terms

For node j, let $T_{k,i}^{(j)}$ be the moment-matched covariance of the ith Bernoulli component. The covariance-quality score is defined as

$$q_{\text{cov},k}^{(j)} = \frac{1}{\epsilon + \frac{1}{M_k}\sum_{i=1}^{M_k} \operatorname{tr}\left(T_{k,i}^{(j)}\right)}, \tag{14}$$

where $\epsilon > 0$ is a small numerical constant used to avoid division by zero. A node with smaller posterior spread therefore receives a larger score.

The link-quality term is derived from realized communication outcomes rather than nominal topology. Let $d_k^{(j)}$ and $\ell_k^{(j)}$ denote the counts of delivered and dropped measurements, respectively. The realized link-quality score is defined as

$$q_{\text{link},k}^{(j)} = \frac{d_k^{(j)}}{d_k^{(j)} + \ell_k^{(j)}}, \tag{15}$$

which directly reflects how reliably that node's information is reaching its neighbors.

Design rationale. The covariance cue is a local posterior-concentration proxy: a smaller average trace means that, after moment projection, the neighbor contributes a sharper Gaussian component to the canonical-form spatial average above. The link cue is deliberately based on delivered and dropped measurements at the current time step rather than on the static graph, because a connected neighbor whose packets are not delivered should not be treated the same as one whose information actually reaches the fusion node.

4.4. Existence-confidence weighting

To capture the missing existence-related dimension, the method computes a certainty score from Bernoulli existence probabilities. For the ith Bernoulli component at node j,

$$c_{k,i}^{(j)} = \left|2r_{k,i}^{(j)} - 1\right|. \tag{16}$$

The per-node weighted existence confidence is then

$$\bar{c}_k^{(j)} = \frac{\sum_{i=1}^{M_k} r_{k,i}^{(j)} c_{k,i}^{(j)}}{\epsilon + \sum_{i=1}^{M_k} r_{k,i}^{(j)}}. \tag{17}$$

The final existence-confidence score is mapped into a bounded positive factor:

$$q_{\text{exist},k}^{(j)} = \lambda_{\min} + (1 - \lambda_{\min})\left(\bar{c}_k^{(j)}\right)^{p_e}, \tag{18}$$

where $\lambda_{\min} \in (0, 1]$ and $p_e > 0$ are tunable parameters. Figure 3 visualizes this bounded Bernoulli-level mapping and representative node-level aggregation effects.

Remark 2 (existence confidence). The mapping $|2r-1|$ assigns low confidence to an undecided Bernoulli component near $r = 0.5$ and high confidence to decisive absence or presence. Weighting this certainty by $r_{k,i}^{(j)}$ makes the node-level score emphasize components that are likely to exist. The lower bound $\lambda_{\min}$ is included because an uncertain existence profile should weaken the existence branch, but it should not by itself erase a neighbor that may still contain useful spatial information.

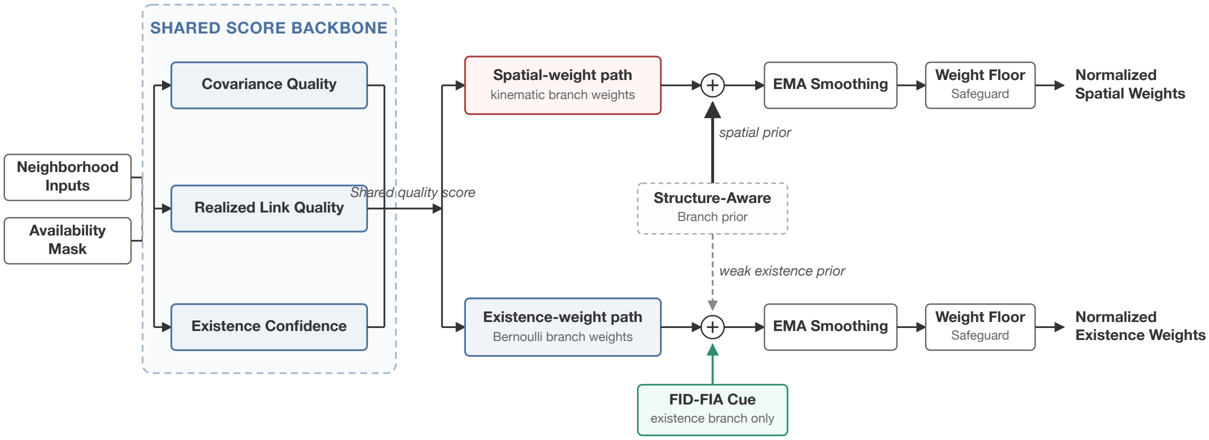

Figure 2: Adaptive fusion-weight factorization, showing the shared quality backbone, decoupled spatial and existence-weight paths, weak structure-aware branch prior, and an FID-FIA cue applied only to the existence branch.

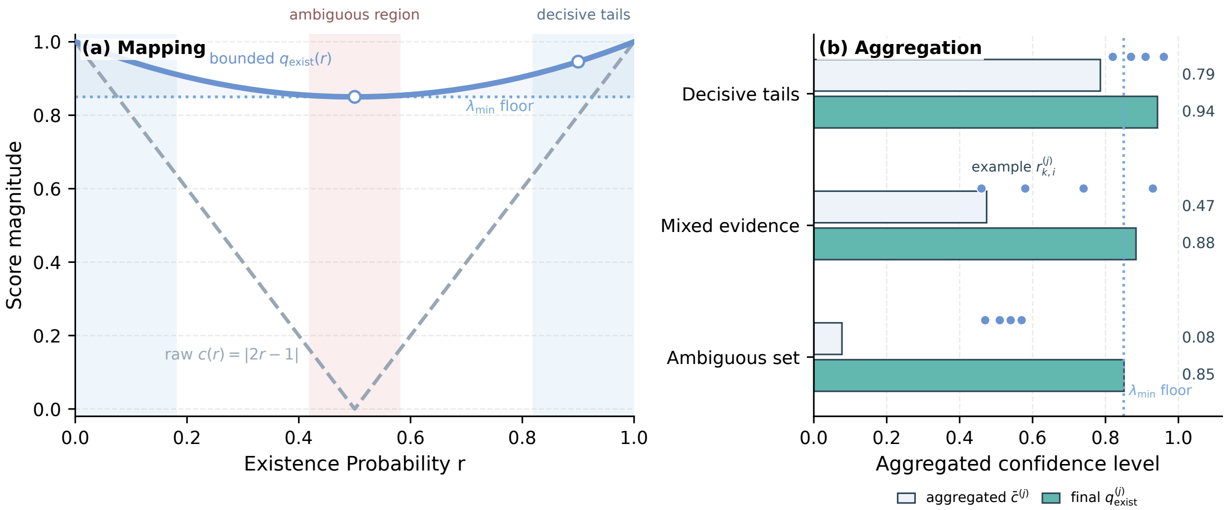

Figure 3: Existence-confidence weighting used in the adaptive rule: (a) the Bernoulli-level decisiveness mapping and bounded score curve, and (b) representative node-level aggregation examples showing how decisive posterior sets produce larger $\bar{c}^{(j)}$ and $q_{\text{exist}}^{(j)}$.

4.5. Decoupled spatial and existence scores

The proposed method does not force the same adaptive score to govern spatial fusion and existence fusion. This separation is structurally motivated by the Bernoulli/LMB form itself: existence and spatial terms enter the fused posterior through different update blocks, even though the formal variational interpretation is deferred to Appendix C. The method therefore constructs branch-specific dedicated scores:

$$\tilde{\omega}_{k,s,\text{sp}}^{(j)} = m_k^{(j)} \cdot \left(q_{\text{cov},k}^{(j)}\right)^{\alpha_x} \cdot \left(q_{\text{link},k}^{(j)}\right)^{\beta_x}, \tag{19}$$

$$\tilde{\omega}^{(j)}_{k,s,\mathrm{ex}} = m^{(j)}_k \cdot \left(q^{(j)}_{\mathrm{link},k}\right)^{\beta_r} \cdot \left(q^{(j)}_{\mathrm{exist},k}\right)^{\alpha_r}. \tag{20}$$

The powers $\alpha_x, \beta_x, \beta_r, \alpha_r \geq 0$ tune the influence of the branch-relevant cues. The final branch scores are obtained through geometric interpolation:

$$\tilde{\omega}^{x,(j)}_{k,s} = \left(\tilde{\omega}^{(j)}_{k,s}\right)^{1-\eta_x} \left(\tilde{\omega}^{(j)}_{k,s,\mathrm{sp}}\right)^{\eta_x}, \qquad \tilde{\omega}^{r,(j)}_{k,s} = \left(\tilde{\omega}^{(j)}_{k,s}\right)^{1-\eta_r} \left(\tilde{\omega}^{(j)}_{k,s,\mathrm{ex}}\right)^{\eta_r}. \tag{21}$$

Here $\eta_x, \eta_r \in [0, 1]$ are interpolation strengths: zero keeps the shared backbone, and one uses the dedicated branch score. The point of this decoupling is not to create two unrelated methods, but to acknowledge that spatial consensus and existence consensus respond differently to weight perturbations. It also determines how Fisher-information-based cues are used in this paper. The information-geometric construction of Cao and Zhao is a natural scalar indicator of local target-separability and posterior informativeness, and in our experiments it is especially effective for cardinality and existence decisions [17]. Related multi-rate average-fusion work also supports Fisher information as a principled local information measure after time alignment [33]. However, applying the same Fisher-type scalar weight to the whole fused posterior can trade away spatial RMSE in the present GA-LMB setting. In the log-utility view developed in Appendix C, a single FID-FIA scalar-weight baseline is a constrained case that forces the same Fisher-separability utility into both the Gaussian spatial barycenter and the Bernoulli existence pool. The branch-decoupled formulation relaxes this constraint: the communication-aware covariance/link-quality branch continues to govern spatial fusion, while the FID-FIA signal is reserved for the existence branch where its empirical strength is most aligned with the fused Bernoulli variable.

Remark 3 (why geometric interpolation). The shared score remains an anchor for both branches, so the spatial and existence paths do not drift into unrelated weight rules. The dedicated spatial and existence scores then tilt the anchor toward the evidence most relevant to each branch. Geometric interpolation preserves the multiplicative reliability interpretation: a weak cue continues to constrain the final score instead of being cancelled by a large additive term.

4.6. FID-FIA existence refinement

The Cardinality-critical mode therefore adds an information-geometric refinement only to the existence branch. This refinement follows the FID-FIA intuition of accumulating pairwise Fisher-information distance between locally represented targets, but it is not used as a replacement scalar fusion rule. For each node j, the method first computes a target-pair accumulation score $a^{(j)}_k$ from the local measurement-updated Bernoulli components. Each Bernoulli component is moment-projected, target pairs are weighted by their existence probabilities, and the pairwise distance is accumulated along the line segment between the two projected states using the local Fisher metric induced by the sensor's detection probability and measurement covariance. The score is then normalized over the available local neighborhood,

$$\hat{a}^{(j)}_{k,s} = \frac{a^{(j)}_k}{\epsilon + \max_{u \in \mathcal{N}_s} a^{(u)}_k}, \qquad 0 \leq \hat{a}^{(j)}_{k,s} \leq 1. \tag{22}$$

This normalized accumulation is mapped into a bounded existence-only modulation factor,

$$q^{(j)}_{\mathrm{fid},k,s} = \rho_{\min} + (1 - \rho_{\min}) \left(\hat{a}^{(j)}_{k,s}\right)^{p_f}. \tag{23}$$

Here $\rho_{\min} \in [0, 1]$ is the lower bound of the FID-FIA existence cue and $p_f > 0$ controls the nonlinearity of the normalized accumulation. Only the existence score is modified:

$$\tilde{\omega}^{r,(j)}_{k,s} \leftarrow \tilde{\omega}^{r,(j)}_{k,s} \left(q^{(j)}_{\mathrm{fid},k,s}\right)^{\gamma_f}. \tag{24}$$

Here $\gamma_f \geq 0$ controls the strength of the FID-FIA modulation. The spatial score $\tilde{\omega}^{x,(j)}_{k,s}$ is left unchanged. In the main experiment, this branch uses a strong existence-only modulation and disables temporal smoothing and minimum-weight flooring for the existence weights, while leaving the spatial branch's structure-aware stabilized weights unchanged. Equivalently, for positive $q^{(j)}_{\mathrm{fid},k,s}$, the update adds $\gamma_f \log q^{(j)}_{\mathrm{fid},k,s}$ to the existence-branch utility and adds nothing to the spatial utility; the zero-floor setting used in the main experiment is the boundary case that permits a zero instantaneous existence weight.

The resulting per-node update is summarized in Algorithm 1. The algorithm is intentionally written at the level of quantities used by the implemented fusion layer: the spatial weights are consumed by the Gaussian canonical-form update, while the existence weights are consumed by the Bernoulli existence update.

Algorithm 1. Communication-aware branch-decoupled GA-LMB fusion at node s and time k.

1. Receive the measurement-updated LMB posteriors $\{\pi^{(j)}_k : j \in \mathcal{N}_s\}$ and the realized delivery statistics for the current step.
2. Form the availability mask $m^{(j)}_k$, covariance score $q^{(j)}_{\mathrm{cov},k}$, link score $q^{(j)}_{\mathrm{link},k}$, and existence-confidence score $q^{(j)}_{\mathrm{exist},k}$.
3. Build the shared score $\tilde{\omega}^{(j)}_{k,s}$ and the spatial/existence dedicated scores $\tilde{\omega}^{(j)}_{k,s,\mathrm{sp}}$ and $\tilde{\omega}^{(j)}_{k,s,\mathrm{ex}}$.
4. Interpolate the shared and dedicated scores to obtain $\tilde{\omega}^{x,(j)}_{k,s}$ and $\tilde{\omega}^{r,(j)}_{k,s}$, then apply the weak structure-aware branch prior.
5. For Cardinality-critical mode only, compute $q^{(j)}_{\mathrm{fid},k,s}$ and multiply it into the existence score; leave the spatial score unchanged.
6. Normalize each branch over active neighbors, optionally apply temporal smoothing and a final weight floor, and write the results as $\omega^{x,(j)}_{k,s}$ and $\omega^{r,(j)}_{k,s}$.
7. Fuse each aligned Bernoulli component by using $\omega^{x,(j)}_{k,s}$ in the Gaussian canonical-form update and $\omega^{r,(j)}_{k,s}$ in the Bernoulli existence update.

4.7. Recommended operating modes

The resulting method family supports two practical operating modes rather than a single mandatory setting. The *Balanced mode* uses the three-factor adaptive backbone, decoupled spatial and existence scores, weak structure-aware refinement, and the standard temporal stabilization, but does not apply the final FID-FIA existence modulation. This mode is recommended when spatial accuracy, RMSE stability, runtime, or energy budget is the primary operational concern.

The *Cardinality-critical mode* adds the FID-FIA cue only to the existence branch. This mode is recommended when the cost of target-number disagreement, missed tracks, or false support is higher than a modest local-RMSE and runtime tradeoff. Both modes share the same communication-aware backbone and the same branch-decoupled fusion algebra; they differ only in whether the final information-geometric cue is activated on the Bernoulli existence path. This framing is important because the experiments show a real operating tradeoff: the backbone is often the stronger low-overhead position-oriented choice, whereas the Cardinality-critical mode is the stronger cardinality-consensus choice with substantially higher information-geometric computation.

4.8. Weak structure-aware refinement and stabilization

After branch decoupling and before the final FID-FIA existence modulation, the method applies a weak graph-aware correction through local structure priors derived from neighborhood-overlap similarity and communication-reliability information. The existence-side structure strength is intentionally kept much smaller than the spatial-side structure strength, so topology provides only a secondary regularization signal and does not dominate the core weight assignment.

The raw adaptive weights can fluctuate sharply over time, especially when packet delivery changes abruptly. To improve stability, the normalized weights are smoothed by an exponential moving average and then passed through a minimum-weight safeguard before final renormalization. This stabilization step is pragmatic rather than a novelty claim by itself; its purpose is to preserve the intended quality ordering while preventing unstable weight collapse in a time-varying communication setting.

4.9. Optional extension modules

The evaluated software also supports several exploratory extension modules outside the claimed core method. Their empirical effect is weaker, more coupled, or less stable than the evidence for the main three-factor backbone plus the retained branch-specific refinements, so they are kept out of the main method description and summarized only in Appendix B.

5. Experimental Setup

This section defines the evaluation protocol used to support the paper's claims. The experiments are organized around one main scenario and a small set of supporting studies. The main scenario is the dual-formation eight-sensor distributed GA-LMB tracking problem under tiered heterogeneous packet loss. Supporting studies are used to check whether the observed gains persist under ideal communication, how the method behaves as communication becomes more constrained, and whether several optional modules should remain outside the core method.

5.1. Scenario, protocols, and operating points

The evaluation is intentionally hierarchical. The tiered-loss dual-formation scenario provides the confirmatory evidence for the proposed GA-LMB method family, while the factor ablation in the same scenario identifies which parts of the adaptive-weight design carry the result. Computational cost is measured on this main scenario as a deployment constraint rather than as a post hoc timing note. Two smaller supporting studies then test the boundaries of the claim: an ideal-communication comparison asks whether the refinement is useful without packet loss, and a communication-level sensitivity probe asks whether the same behavior persists as communication constraints become more severe. Additional exploratory variants are treated as appendix material rather than as competing headline methods.

Main scenario. The primary evaluation setting uses eight mobile sensors arranged as two four-sensor formations. The filtering mode is distributed local fusion, meaning that each node fuses only its own posterior and the posteriors available inside its communication neighborhood. The local multi-object filter is GA-LMB with loopy belief propagation (LBP) for approximate marginal data association [38].

The choice of the geometric-average KLA route is deliberate. GA fusion gives a conservative logarithmic-opinion-pool update under unknown cross-correlations, admits a stable canonical-form Gaussian fusion step for the LMB spatial densities, and naturally exposes the Bernoulli existence product rule in which branch-specific weighting can be applied. These properties make it a cleaner vehicle for studying adaptive weight allocation than a broad comparison across fusion families.

The common sensor-side configuration is:

- number of sensors: 8,
- filter mode: GA-LMB,
- data association: LBP,
- clutter rate: 3 for every sensor,
- detection probability: 0.9 for every sensor,
- measurement-noise scale: $q = 3$ for every sensor,
- sensor field of view enabled with half-angle 60° and range 60000.

The nominal detection probability, measurement-noise scale, and clutter rate are therefore homogeneous across sensors. The sensing variation exercised by the main scenario comes from the dual-formation geometry, finite fields of view, target visibility, and the resulting posterior covariance and existence differences; it should not be read as a sensitivity test over explicit p_D, measurement-noise, or clutter tiers.

The target side consists of ten objects arranged in three groups with staggered births. The three groups contain $3 + 3 + 4$ targets, respectively, and the targets move toward the central region with nominal speed 0.45. Births are staggered every eight time steps, and the total simulation length is 100 steps.

Communication model. The main scenario uses a tiered heterogeneous packet-loss setting. Communication is applied before distributed fusion through a measurement-delivery model with bandwidth limitation and link-level packet loss. The main configuration is:

- communication level: 2,
- global maximum delivered measurements per step: 80,

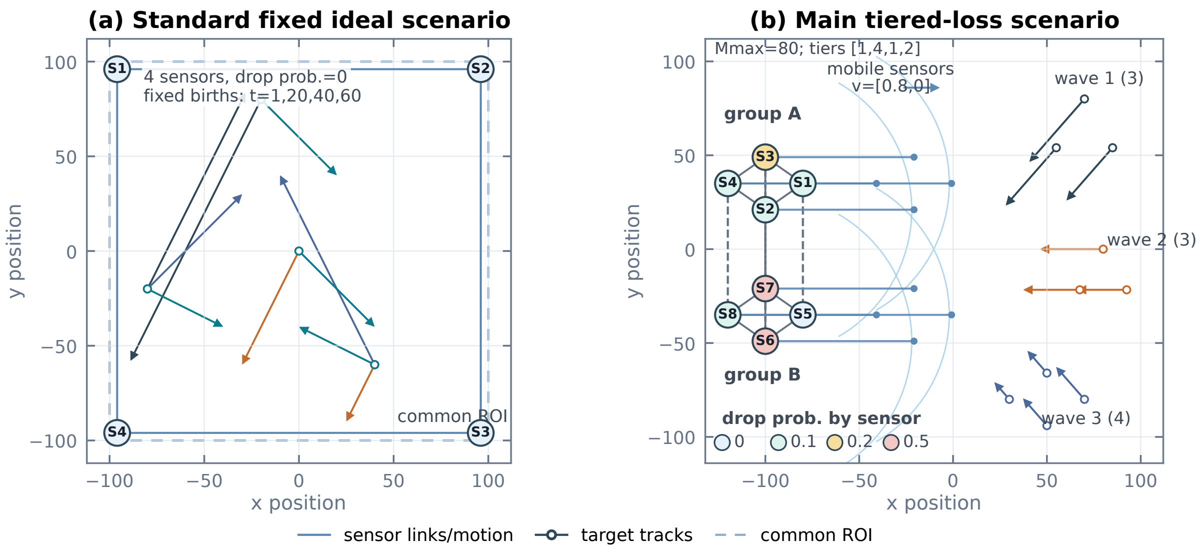

Figure 4: Schematic comparison between the standard fixed ideal benchmark and the main tiered heterogeneous packet-loss scenario. The ideal benchmark uses four full-ROI sensors with no packet loss and fixed target births; the main scenario uses the dual-formation eight-sensor geometry and staggered target waves with the bandwidth cap and persistent drop-tier heterogeneity used in the main experiments.

- packet-selection policy: weighted priority with random measurement selection,
- link model: fixed packet drop,
- target mean drop rate: $p_{\text{drop}} = 0.2$,
- tiered drop levels: $[0, 0.1, 0.2, 0.5]$,
- tiered level counts: $[1, 4, 1, 2]$.

This tiered design preserves the historical mean communication degradation while introducing persistent cross-node heterogeneity.

Figure 4 summarizes the contrast between the simpler standard fixed ideal benchmark and the main tiered-loss scenario. The standard ideal benchmark uses four full region-of-interest (ROI) sensors, ideal communication, and fixed target births. The main scenario is the dual-formation eight-sensor case with staggered target waves, a bandwidth cap, and persistent drop-tier heterogeneity before distributed fusion.

Mainline ablation protocol. The core ablation study compares five consistently named arms:

1. Fixed Metropolis,
2. Covariance-only adaptive,
3. Covariance-link adaptive,
4. Three-factor adaptive backbone,
5. Balanced mode, which adds weak structure-aware modulation after covariance, link-quality, and existence-confidence weighting.

This five-arm path is a design diagnostic for the communication-aware backbone and the pre-refinement branch split. The manuscript results table keeps the visible rows focused on operating points by treating the three-factor backbone as the reliability model used by the branch-decoupled Balanced mode rather than as a separate deployment arm. The main confirmation additionally compares probability-of-detection (PD)-weighted GA, FID-FIA-weighted

GA used as the information-geometric baseline, and the Cardinality-critical mode to show the branch-specific FID-FIA extension on the same scale. The Balanced mode is the position-sensitive operating mode. It uses the same parameterization across all trials: exponential smoothing with $\alpha = 0.7$, minimum weight 0.05, existence-confidence parameters $\lambda_{\min} = 0.85$ and $p_e = 2.0$, branch-decoupling strengths $(0.5, 0.15)$ for the spatial and existence branches, and weak structure-aware refinement strengths $(0.45, 0.08)$. The Cardinality-critical mode keeps the spatial side unchanged but applies FID-FIA information geometry only to the existence weights, with existence-branch smoothing and minimum-weight flooring disabled for that refinement. Exploratory extension modules are disabled in the main configuration.

5.2. Metrics and reporting

The experiments report three categories of metrics. The first category is network-level disagreement among sensors after distributed fusion. These primary metrics are OSPA consensus error, matched localization disagreement, and cardinality dispersion as defined in Section 3. The second category is conventional truth-referenced tracking accuracy. These secondary metrics are local E-OSPA, local RMSE, and local cardinality error. The third category is computational cost, measured as filter/fusion wall-clock runtime per trial, runtime per simulation step, and runtime relative to Fixed Metropolis. The OSPA/GOSPA family is used because it provides mathematically consistent finite-set distances for multi-object filtering and tracking evaluation [34, 35, 37]. The primary disagreement metrics are task-specific network-agreement measures built from these standard ingredients, not a replacement for standard truth-referenced tracking benchmarks, so the paper always interprets them together with the secondary local metrics and the computational-cost measurements rather than in isolation.

This hierarchy is deliberate. The proposed method is not intended as a generic single-sensor accuracy booster. It is designed as a distributed fusion-weight mechanism for reducing disagreement in the fused multi-sensor picture under heterogeneous communication. For that reason, the paper treats the network-level disagreement metrics as the main outcome, the local tracking metrics as a safeguard against hidden degradation, and computational cost as a first-class deployment-feasibility axis. The target is to improve cross-node agreement without causing a meaningful drop in conventional tracking quality or requiring impractical additional computation. Runtime is measured only around the distributed LMB filtering/fusion call for each arm; scenario generation, communication-model sampling, plotting, and metric evaluation are excluded so that the reported cost isolates the algorithmic overhead of adaptive weighting and branch-specific FID-FIA scoring.

Unless otherwise stated, reported scalar metrics are Monte Carlo means. The main fixed-versus-existence-refined confirmation, the five-arm factor ablation, the ideal-communication comparison, and the communication-level sensitivity study all use 50 deterministic paired trials with seeds 2–51. The main confirmation, ideal-communication comparison, and communication-level sensitivity study are paired by construction. The five-arm factor ablation remains primarily a design diagnostic, but it is now reported on the same 50-trial seed set for consistency.

6. Results and Discussion

This section reports the empirical evidence supporting the paper's main claims. The strongest evidence comes from the tiered heterogeneous packet-loss GA scenario introduced in Section 5, so that scenario is treated as the primary result. Ideal communication and communication-sensitivity experiments are used as supporting studies. Appendix-only results document exploratory variants without diluting the main GA/KLA method.

6.1. Main comparison and design ablation

Consensus and local safeguards. The main claim is that communication-aware adaptive weighting substantially improves distributed consensus quality in the eight-sensor dual-formation GA-LMB scenario under tiered heterogeneous packet loss. Because the paper's design target is inter-sensor agreement, this subsection first reports the primary disagreement metrics and then checks the conventional local tracking metrics as a safeguard. The comparison separates three questions: whether a direct reliability-weighted GA baseline is already beneficial, whether the communication-aware branch-decoupled design adds further value, and whether an existence-only information-geometric cue is useful when cardinality errors dominate. The PD-weighted GA row answers the first question by using detection-probability-style reliability weighting.

Table 1 should therefore be read as an ordered pattern rather than as isolated numeric rows. The PD-weighted GA baseline improves over Fixed Metropolis, confirming that adaptive reliability information is useful under heterogeneous

Table 1
Main paired consensus comparison under tiered heterogeneous packet loss. Values are mean ± standard deviation over 50 deterministic paired trials with seeds 2–51. Lower values are better.

Arm	OSPA err.	Loc. disag.	Card. disp.
Fixed Metropolis	2.469 ± 0.220	2.326 ± 0.353	0.716 ± 0.224
PD-weighted GA	2.177 ± 0.161	1.995 ± 0.233	0.588 ± 0.170
FID-FIA-weighted GA	1.818 ± 0.056	1.643 ± 0.109	0.123 ± 0.023
Balanced mode	1.779 ± 0.073	$\mathbf{1.522} \pm 0.157$	0.188 ± 0.031
Cardinality-critical mode	$\mathbf{1.678} \pm 0.050$	1.546 ± 0.158	$\mathbf{0.063} \pm 0.017$

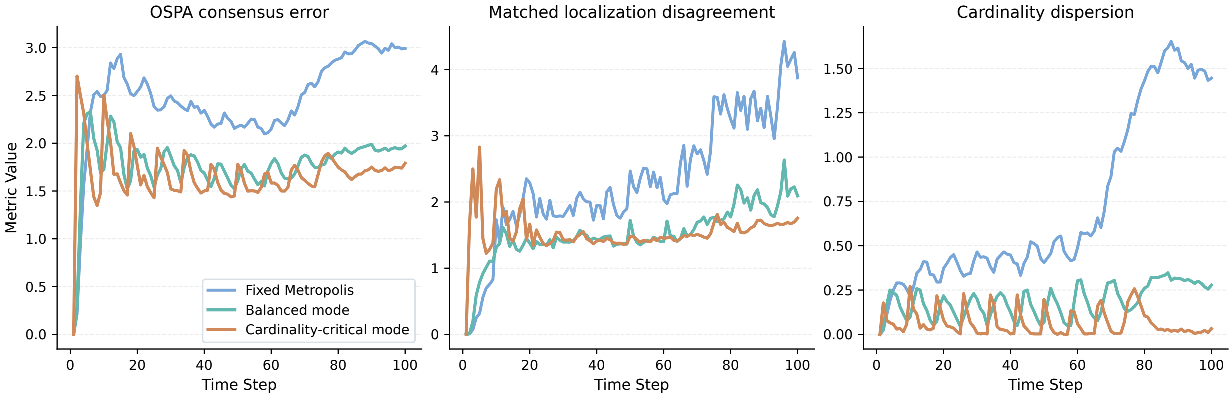

Figure 5: Mean OSPA consensus error, matched localization disagreement, and cardinality dispersion over time for Fixed Metropolis, Balanced mode, and Cardinality-critical mode under tiered heterogeneous packet loss. Curves are 50-trial means over deterministic paired trials with seeds 2–51.

delivery. The information-geometric FID-FIA-weighted GA row is especially strong for cardinality dispersion. The communication-aware method family, however, provides the stronger overall consensus profile. The Balanced mode gives the best localization-oriented disagreement result, while the Cardinality-critical mode gives the lowest OSPA consensus error and cardinality dispersion: relative to Fixed Metropolis, it reduces OSPA consensus error from 2.469 to 1.678, matched localization disagreement from 2.326 to 1.546, and cardinality dispersion from 0.716 to 0.063. These correspond to reductions of 32.1%, 33.5%, and 91.2%, respectively. For the original method-family arms, the paired reductions relative to Fixed Metropolis are positive in all 50 trials for all three primary disagreement metrics, with exact sign-test $p = 1.8 \times 10^{-15}$. Figure 5 plots the 50-trial mean of the per-time-step disagreement trajectories without additional temporal smoothing; both operating modes stay below the fixed baseline across most of the trajectory, with the Cardinality-critical mode most clearly separating on cardinality dispersion.

The primary-metric gain would be less convincing if it came at the cost of clear local tracking degradation. Table 2 provides this safeguard check. The PD-weighted GA baseline attains the lowest truth-referenced local RMSE among the retained rows, but it remains substantially weaker on local E-OSPA and local cardinality error. The Cardinality-critical mode shows the complementary behavior: it gives the best local E-OSPA and local cardinality error, while its RMSE is higher than the PD-weighted GA, Fixed Metropolis, and Balanced rows but still slightly lower than FID-FIA-weighted GA. Thus the large cardinality-dispersion gain is not simply agreement around an incorrect target count. It is better understood as the effect of injecting information-geometric evidence into the existence branch while retaining the spatial-consensus advantages of the communication-aware adaptive backbone.

The two proposed operating-mode rows should therefore be read as recommended settings of the same method family rather than as an inconsistency in the proposed design. The Balanced mode is preferable when spatial RMSE stability is the dominant requirement. The Cardinality-critical mode is preferable when target-number agreement and truth-referenced cardinality error are the dominant risks. This usage-oriented distinction makes the tradeoff explicit instead of hiding it behind a single headline configuration.

Table 2
Conventional local tracking metrics in the main tiered heterogeneous packet-loss scenario. Values are mean ± standard deviation across the same 50 deterministic paired trials used in Table 1, after averaging each trial over sensors.

Arm	E-OSPA	RMSE	CardErr
Fixed Metropolis	2.863 ± 0.124	1.650 ± 0.078	1.455 ± 0.242
PD-weighted GA	2.736 ± 0.116	$\mathbf{1.563} \pm 0.061$	1.255 ± 0.205
FID-FIA-weighted GA	2.185 ± 0.050	1.734 ± 0.094	0.388 ± 0.047
Balanced mode	2.335 ± 0.072	1.606 ± 0.050	0.579 ± 0.067
Cardinality-critical mode	$\mathbf{2.020} \pm 0.047$	1.721 ± 0.161	$\mathbf{0.224} \pm 0.029$

Table 3
Computational cost in the main tiered heterogeneous packet-loss scenario. Wall-clock time measures only the distributed LMB filtering/fusion call for each arm over 50 deterministic paired trials with seeds 2–51. The Fixed Metropolis, FID-FIA-weighted GA, Balanced, and Cardinality-critical rows come from the operating-mode runtime probe; the PD-weighted GA row comes from the direct dynamic-weighting probe under the same scenario and seed set. Relative runtime is computed against the paired Fixed Metropolis denominator within the corresponding probe.

Arm	Runtime (s)	Runtime/step (s)	Relative runtime
Fixed Metropolis	52.123 ± 7.932	0.521	1.000×
PD-weighted GA	61.668 ± 5.376	0.617	1.233×
FID-FIA-weighted GA	147.674 ± 23.957	1.477	2.833×
Balanced mode	56.378 ± 9.626	0.564	1.082×
Cardinality-critical mode	155.913 ± 18.220	1.559	2.991×

Computational cost is the third evaluation axis for this method family. Table 3 reports 50-trial runtime measurements in the same 100-step tiered-drop scenario and covers all arms in the main comparison. The measurement includes only the distributed LMB filtering/fusion call for each arm; scenario generation, communication-model sampling, plotting, and metric evaluation are excluded. The main cost distinction is clear. The Balanced mode remains close to the fixed-weight baseline, requiring about 1.08× the Fixed Metropolis runtime. The PD-weighted GA baseline is also a moderate-cost dynamic-weighting alternative at about 1.23× fixed runtime. By contrast, FID-FIA-weighted GA and the Cardinality-critical mode are both near the three-times-cost regime. The runtime evidence therefore identifies FID-FIA scoring, not the basic communication-aware weighting backbone or the direct PD-weighted baseline, as the dominant source of additional cost.

The cost results sharpen the operating-mode recommendation. The Balanced mode is the low-overhead default when spatial consensus, RMSE stability, runtime, or energy budget is the limiting requirement. The Cardinality-critical mode should be selected when target-number agreement and suppression of missed or false support justify roughly three times the fixed-weight filtering/fusion time.

Factor ablation. The most important diagnostic ablation is the factor-by-factor path from Fixed Metropolis to the Balanced mode. Table 4 summarizes this 50-trial path and adds the Cardinality-critical mode as the last row. To keep the table focused on the main operating points, the three-factor adaptive backbone is not listed as a separate row; it remains the reliability model used by the branch-decoupled Balanced mode. The last row is the branch-specific existence refinement that converts the Balanced mode into the cardinality-sensitive mode.

Several conclusions follow from Table 4. First, covariance weighting is necessary but not sufficient. It improves over fixed topology weights, confirming that posterior concentration is a useful first-order proxy for local posterior quality, but it still leaves a substantial gap to the branch-decoupled operating modes.

Second, realized link quality is the dominant communication-aware factor under heterogeneous packet loss. Adding it after covariance produces the largest incremental change in the backbone path, especially for cardinality dispersion, which is approximately halved. This is the key evidence that the method is not merely reweighting sharper posteriors; it is also responding to whether those posteriors are reliably delivered to neighboring nodes.

Third, existence confidence remains part of the adaptive backbone, but its effect is not isolated as a separate operating row in Table 4. The reason is methodological rather than negative: existence confidence is used together

Table 4
Network disagreement metrics for the main factor ablation under tiered heterogeneous packet loss. Values are mean ± standard deviation over 50 deterministic paired trials. The rows show the progression from fixed weighting to the covariance-link backbone and then to the Balanced operating mode, which includes the three-factor adaptive backbone and weak structure-aware branch decoupling. The final row applies the FID-FIA cue only to the existence-weight path to obtain the cardinality-critical operating mode.

Arm	OSPA err.	Loc. disag.	Card. disp.
Fixed Metropolis	2.469 ± 0.220	2.326 ± 0.353	0.716 ± 0.224
Covariance-only adaptive	2.091 ± 0.145	1.951 ± 0.253	0.466 ± 0.134
Covariance-link adaptive	1.788 ± 0.071	1.525 ± 0.156	0.188 ± 0.031
Balanced mode	1.779 ± 0.073	$\mathbf{1.522} \pm 0.157$	0.188 ± 0.031
Cardinality-critical mode	$\mathbf{1.678} \pm 0.050$	1.546 ± 0.158	$\mathbf{0.063} \pm 0.017$

with branch decoupling in the Balanced mode, where the reliability model is applied differently to the spatial and Bernoulli existence paths. This is consistent with the design rationale: covariance and link quality do not directly express how decisive a local Bernoulli component is about target existence.

Finally, the Balanced mode provides the pre-FID-FIA branch-decoupled operating point. The final row changes the operating point by applying the FID-FIA cue only to the existence branch, reducing cardinality dispersion from the approximately 0.188 level of the Balanced/backbone setting to 0.063 while also lowering OSPA consensus error. This is why the proposed method is presented as a configurable branch-decoupled family: the Balanced mode is the spatial/position-oriented setting, and the Cardinality-critical mode is the cardinality-oriented setting.

6.2. Supporting communication studies

Ideal communication. An obvious question is whether the weak structure-aware decoupled refinement only helps because it compensates for packet loss. The ideal-communication experiment addresses this point by setting the communication level to zero and comparing Fixed Metropolis, PD-weighted GA, FID-FIA-weighted GA, the Balanced mode, and the Cardinality-critical mode under otherwise matched conditions. Table 5 reports this comparison over the same 50 deterministic seed set used in the main confirmation.

Table 5
Ideal-communication supporting comparison over 50 deterministic paired trials with seeds 2–51. Adaptive rows report the mean value and relative change versus Fixed Metropolis; lower values are better, so negative percentages indicate improvements.

	Network disagreement			Local safeguards	
Arm	OSPA err.	Loc. disag.	Card. disp.	E-OSPA	RMSE
Fixed Metropolis	1.634	1.381	0.090	1.908	1.442
PD-weighted GA	1.434 (-12.2%)	1.218 (-11.8%)	0.056 (-37.6%)	1.825 (-4.3%)	1.377 (-4.5%)
FID-FIA-weighted GA	1.534 (-6.1%)	1.333 (-3.5%)	0.057 (-36.3%)	1.881 (-1.4%)	1.500 (+4.0%)
Balanced mode	**1.427 (-12.7%)**	**1.202 (-13.0%)**	0.071 (-20.9%)	1.839 (-3.6%)	**1.375 (-4.7%)**
Cardinality-critical mode	1.433 (-12.3%)	1.303 (-5.7%)	**0.049 (-46.0%)**	**1.766 (-7.4%)**	1.470 (+2.0%)

The ideal-communication result mirrors the branch specialization observed in the tiered-drop experiment. The PD-weighted GA row confirms that a simple reliability-weighted GA baseline can also help when link degradation is removed. Information-geometric weighting is especially effective for cardinality dispersion, whereas the Balanced mode remains the localization-oriented alternative and gives the lowest OSPA consensus error, matched localization disagreement, and local RMSE in this benign setting. The Cardinality-critical mode gives the lowest cardinality dispersion and local E-OSPA without forcing a single FID-FIA weight onto the whole posterior. This supporting study therefore serves two purposes: it shows that the refinement is not merely repairing packet loss, and it shows that the consensus gain is not purchased by a collapse in local tracking quality.

Communication sensitivity and robustness. A communication-level sensitivity study was added to test whether the two operating modes remain useful outside the single tiered-drop setting. The study uses the same eight-sensor dual-formation GA-LMB scenario and compares Fixed Metropolis, Balanced mode, and Cardinality-critical mode over 50 deterministic paired trials at each communication level. Level 0 is ideal communication, level 1 adds only the

Table 6
Communication-level sensitivity in the eight-sensor dual-formation GA-LMB scenario. Each level uses 50 deterministic paired trials. Values are Fixed Metropolis / Balanced mode / Cardinality-critical mode, and lower values are better.

Level	Additional constraint	OSPA err.	Loc. disag.	Card. disp.
0	none	1.628/1.423/1.432	1.378/1.201/1.303	0.087/0.069/0.048
1	bandwidth cap	1.701/1.482/1.473	1.447/1.242/1.344	0.126/0.085/0.049
2	tiered link loss	2.383/1.760/1.669	2.263/1.536/1.571	0.650/0.180/0.066
3	node outage	2.656/1.808/1.689	2.701/1.570/1.574	0.880/0.201/0.066

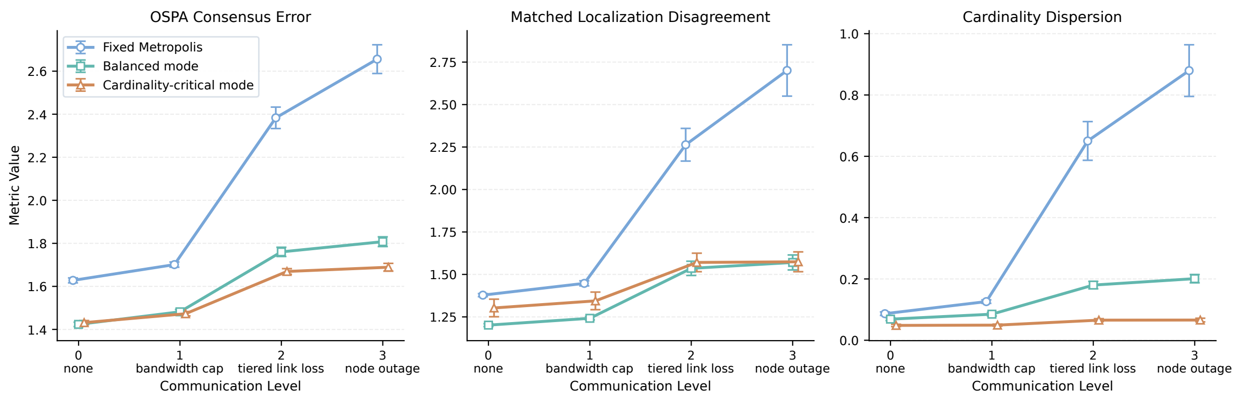

Figure 6: Communication-level sensitivity for Fixed Metropolis, Balanced mode, and Cardinality-critical mode. Markers show 50-trial means and error bars show 95% confidence intervals.

global bandwidth cap, level 2 adds the tiered fixed link-loss model used in the main experiment, and level 3 further adds random node outages. Table 6 reports the resulting network-level disagreement metrics, and Figure 6 visualizes the same means with 95% confidence intervals. This table is a communication-sensitivity check; it does not test explicit sensor-side p_D or measurement-noise tiers.

This study is not used to replace the main tiered-drop comparison, but it closes the previous gap in the communication-sensitivity evidence. Both proposed operating modes improve on Fixed Metropolis at every communication level. The important pattern is the gradient of the effect: the gain is modest under ideal or bandwidth-only communication, but it becomes much larger once tiered link loss or node outages make realized delivery quality heterogeneous. The Balanced mode gives the lowest matched localization disagreement at every level, while the Cardinality-critical mode gives the lowest cardinality dispersion at every level and the lowest OSPA consensus error once communication constraints are nontrivial. The local safeguard metrics move in the same direction: local E-OSPA and local cardinality error are lowest for the Cardinality-critical mode, while local RMSE remains strongest for the Balanced mode. This supports the interpretation that communication-aware weighting is most valuable when communication quality varies across nodes, and that the two operating modes expose a meaningful spatial-versus-cardinality tradeoff rather than a single brittle tuning.

7. Conclusion

This paper studies communication-aware adaptive weights for distributed geometric-average KLA-based LMB fusion under heterogeneous packet delivery. The main result is deliberately focused: fixed topology weights are too rigid when posterior concentration, Bernoulli existence decisiveness, and realized communication quality vary across nodes. A factorized backbone using covariance quality, delivered-link quality, and existence confidence gives a low-overhead way to adapt the GA-LMB fusion weights, while branch decoupling lets the spatial and existence parts respond to different reliability cues.

The experiments support this operating-mode view. In the eight-sensor tiered-loss scenario, both proposed modes improve network-level disagreement relative to Fixed Metropolis and dynamic-weight baselines. Balanced mode is the position-sensitive choice: it keeps runtime close to fixed weighting and gives the strongest localization-oriented behavior. Cardinality-critical mode injects the FID-FIA cue only into the existence branch, giving the strongest target-number agreement and local cardinality safeguards at roughly the cost of other FID-FIA-based fusion. The ideal-communication and communication-level studies show that the effect is not only packet-loss compensation, but the gain is largest when realized delivery becomes heterogeneous.

The present evidence should be read within its intended scope. The reported scenario exercises geometry/FOV-induced sensing variation and communication heterogeneity, while nominal detection probability, clutter rate, and measurement-noise scale remain homogeneous across sensors. Future work should therefore test richer heterogeneous sensing, stronger external baselines, larger Monte Carlo suites, lower-cost FID-FIA implementations, and asynchronous or multi-rate distributed LMB fusion where availability can be replaced by time-aligned posterior propagation and information decay.

A. Simulation Setup

Section 5 defines the common filter, sensor, communication, operating-mode, and Monte Carlo settings. The remaining constants needed to reproduce the scenario are the formation geometry, target initialization, and paired-run handling below.

The eight sensors form two four-sensor groups. Their formation centers start at $[-80, 35]^{\mathsf{T}}$ and $[-80, -35]^{\mathsf{T}}$ and move with nominal velocity $[0.8, 0]^{\mathsf{T}}$. Each group follows a four-agent leader-follower pattern with spacing 20. The communication graph is fully connected inside each group, with cross links $1 \leftrightarrow 5$, $2 \leftrightarrow 6$, $3 \leftrightarrow 7$, and $4 \leftrightarrow 8$; the communication range is 150.

The ten targets are initialized as the three staggered groups summarized in Section 5. Their group centers are $[70, 80]^{\mathsf{T}}$, $[80, 0]^{\mathsf{T}}$, and $[70, -80]^{\mathsf{T}}$, with spacings 30, 25, and 20, respectively. Births start at time step one, repeat every eight steps, and last through the 100-step scenario.

Within each deterministic seed, all compared arms share the same generated truth, measurements, communication realization, and delivered measurement sets. The ideal-communication comparison reuses the same geometry and targets but sets communication level 0, uses effectively unlimited delivery, and sets $p_{\text{drop}} = 0$. The appendix-only AA diagnostics use the same communication-constrained scenario, with a shorter AA horizon and stronger AA-specific pruning to control mixture growth.

B. Additional Attempted Modules and Results

The secondary experiments below are boundary checks rather than extra components of the method. Unless noted, they reuse the tiered-loss scenario and reporting protocol from Section 5.

AA-based secondary route. An arithmetic-average (AA) diagnostic checks whether the same reliability cues transfer to a different fusion operator. The AA run uses the horizon and pruning changes noted in Appendix A; its merger consumes the weights through linear Bernoulli averaging and existence-weighted Gaussian-mixture concatenation rather than GA/KLA logarithmic pooling.

Table 7
Appendix-only AA-based secondary route.

Arm	OSPA	Loc. disag.	Cardinality
fixed AA	3.903	5.130	0.226
covariance-link AA	3.507	4.401	0.103
Balanced AA	3.449	4.306	0.102
Cardinality-critical AA	3.457	4.328	0.095

Balanced AA reduces OSPA consensus error, matched localization disagreement, and cardinality dispersion by about 11.6%, 16.1%, and 55.0% relative to fixed AA. Cardinality-critical AA gives the lowest cardinality dispersion, about a 58.0% reduction, but is slightly worse than Balanced AA on OSPA and localization disagreement. The pattern matches the main operating-mode tradeoff, while the evidence remains secondary because the fusion operator is different.

Other exploratory modules. Several exploratory modules were also left outside the core rule because their effects were weak, mixed, or negative:

Table 8
Exploratory modules not retained in the main method.

Module	Observed effect	Decision
NIS consistency weighting	Plain NIS worsens all three disagreement metrics; robust NIS remains close to the no-NIS baseline but does not improve OSPA or cardinality dispersion.	Exclude from core rule.
Freshness and history weighting	Freshness is effectively flat; history gives only a small localization change while OSPA and cardinality are unchanged or slightly worse.	No main-text role.
Direct cardinality-consensus score	Pilot degrades OSPA, RMSE, and cardinality dispersion relative to the covariance-link baseline.	Negative appendix evidence.
Association ambiguity / stronger posterior structure	Association ambiguity is mixed; stronger posterior-structure consistency ties the retained weak static structure prior.	Future extension candidate.

These results keep the design space visible without turning every tested cue into a method component. The retained method stays focused on covariance quality, realized link quality, existence confidence, branch decoupling, and the existence-branch FID-FIA cue.

C. Theoretical Interpretation of Adaptive Branch-Decoupled Fusion

The proof chain behind the adaptive branch-decoupled rule is deliberately limited. It assumes the same label alignment and moment-compatible Gaussian projection used by the implemented merger, together with standard geometric-average KLA and LMB background [5–7, 12–15]. It should not be read as a global optimality theory for unconstrained LMB fusion.

C.1. Entropy-Regularized Interpretation of the Weight Backbone

Fix a fusion node s at time k with active neighborhood $\mathcal{A}_{k,s}$. The adaptive backbone constructs the positive score

$$s_{0,j} = q^{(j)}_{\mathrm{cov},k} q^{(j)}_{\mathrm{link},k} q^{(j)}_{\mathrm{exist},k}, \qquad j \in \mathcal{A}_{k,s}, \tag{25}$$

where the three terms denote covariance quality, realized link quality, and existence confidence. Define $u_{0,j} = \log s_{0,j}$ and

$$\Delta_{k,s} = \left\{ w \in \mathbb{R}^{|\mathcal{A}_{k,s}|} : w_j \geq 0, \sum_j w_j = 1 \right\}. \tag{26}$$

Proposition C1. For any $\tau > 0$, the program

$$\max_{w\in\Delta_{k,s}} \left\{ \sum_j w_j u_{0,j} + \tau H(w) \right\}, \qquad H(w) = -\sum_j w_j \log w_j, \tag{27}$$

has the softmax solution

$$w^\star_j = \frac{\exp(u_{0,j}/\tau)}{\sum_u \exp(u_{0,u}/\tau)} = \frac{s_{0,j}^{1/\tau}}{\sum_u s_{0,u}^{1/\tau}}. \tag{28}$$

Thus $\tau = 1$ recovers the implemented normalized product rule. The multiplicative backbone is an entropy-regularized allocation in log-utility space.

Covariance score in the isotropic limit. If the local Gaussian components satisfy $T^{(j)}_{k,i} = \sigma_j^2 I_d$, then the implemented covariance score

$$q^{(j)}_{\mathrm{cov},k} = \frac{1}{\epsilon + d\sigma_j^2} \tag{29}$$

induces the same ordering as smaller σ_j^2, smaller $\log\det T^{(j)}_{k,i}$, larger $\log\det(T^{(j)}_{k,i})^{-1}$, and larger $\mathrm{tr}((T^{(j)}_{k,i})^{-1})$. The trace-based score therefore agrees with standard information criteria in the isotropic case; near-isotropic differences enter through anisotropy.

C.2. Algebraic Branch Structure of the Implemented GA-LMB Merger

The implemented merger uses separate spatial and existence weights because the GA-LMB update already exposes two different algebraic blocks.

Spatial branch. For a fixed Bernoulli component i, the Gaussian canonical fusion step is

$$K_i = \sum_j w_{x,j} T_{i,j}^{-1}, \tag{30}$$

$$h_i = \sum_j w_{x,j} T_{i,j}^{-1} v_{i,j}, \tag{31}$$

with $\Sigma_i = K_i^{-1}$ and $\mu_i = \Sigma_i h_i$.

Proposition C2. Conditional on w_x, $\mathcal{N}(\mu_i, \Sigma_i)$ is the minimizer of

$$\min_{q\in\mathcal{G}} \sum_j w_{x,j} D_{\mathrm{KL}}\left(q \,\|\, \mathcal{N}(v_{i,j}, T_{i,j})\right), \tag{32}$$

where $\mathcal{G}$ is the Gaussian family. Expanding the Gaussian KL in canonical parameters gives the precision and information-vector averages above, so the spatial update is the reverse-KL Gaussian barycenter.

Existence branch. The same merger computes

$$r_i = \frac{\eta_i(w_x) \prod_j r_{i,j}^{w_{r,j}}}{\eta_i(w_x) \prod_j r_{i,j}^{w_{r,j}} + \prod_j (1 - r_{i,j})^{w_{r,j}}}, \tag{33}$$

where $\eta_i(w_x)$ is the spatial-overlap normalization induced by the Gaussian branch.

Proposition C3. The fused existence probability satisfies

$$\log \frac{r_i}{1 - r_i} = \log \eta_i(w_x) + \sum_j w_{r,j} \log \frac{r_{i,j}}{1 - r_{i,j}}. \tag{34}$$

Thus, for fixed spatial weights, the existence branch is a weighted Bernoulli logit pool with an additive spatial-overlap offset.

C.3. From Bernoulli-RFS Decomposition to an Aligned-LMB Surrogate

Proposition C4. For Bernoulli RFS densities $\beta = (r, p)$ and $\beta_j = (r_j, p_j)$,

$$D_{\mathrm{KL}}(\beta \,\|\, \beta_j) = d_{\mathrm{Ber}}(r \,\|\, r_j) + r\, D_{\mathrm{KL}}(p \,\|\, p_j), \tag{35}$$

where

$$d_{\mathrm{Ber}}(r \,\|\, r_j) = (1 - r) \log \frac{1 - r}{1 - r_j} + r \log \frac{r}{r_j}. \tag{36}$$

The spatial mismatch is therefore naturally weighted by the existence level.

Proposition C5. For a fixed Bernoulli component i, consider the decoupled surrogate

$$\mathcal{J}_i(r, p; w_x, w_r) = \sum_j w_{r,j}\, d_{\mathrm{Ber}}(r \,\|\, r_{i,j}) + r \sum_j w_{x,j}\, D_{\mathrm{KL}}(p \,\|\, p_{i,j}). \tag{37}$$

Its minimizer satisfies

$$p_i^{\star}(x) = \frac{1}{\eta_i(w_x)} \prod_j p_{i,j}(x)^{w_{x,j}}, \tag{38}$$

$$\log \frac{r_i^\star}{1-r_i^\star} = \log \eta_i(w_x) + \sum_j w_{r,j} \log \frac{r_{i,j}}{1-r_{i,j}}. \tag{39}$$

The spatial equation follows by normalizing the weighted geometric product. Substituting this optimum gives the $-\log \eta_i(w_x)$ spatial contribution, and differentiating the Bernoulli part gives the logit identity. With aligned labels and product LMB form, the same argument lifts labelwise because the reverse-KL divergence is additive across aligned Bernoulli factors.

Scope boundary. This surrogate does not make the branch-decoupled rule the unique optimizer of a fully unconstrained LMB fusion objective. It only states that the implemented update is compatible with a per-Bernoulli aligned-LMB objective under the merger's label-alignment and projection assumptions.

C.4. Branch-Decoupled Weight Allocation and Structure-Aware Regularization

The branch-specific scores are

$$s_{x,j} = \left(q_{\mathrm{cov},j}\right)^{\alpha_x} \left(q_{\mathrm{link},j}\right)^{\beta_x}, \tag{40}$$

$$s_{r,j} = \left(q_{\mathrm{link},j}\right)^{\beta_r} \left(q_{\mathrm{exist},j}\right)^{\alpha_r}, \tag{41}$$

with geometric interpolation

$$\tilde{s}_{x,j} = s_{0,j}^{1-\eta_x} s_{x,j}^{\eta_x}, \tag{42}$$

$$\tilde{s}_{r,j} = s_{0,j}^{1-\eta_r} s_{r,j}^{\eta_r}. \tag{43}$$

This is convex interpolation in log-utility space. Define $\hat{u}_{x,j} = \log \tilde{s}_{x,j}$ and $\hat{u}_{r,j} = \log \tilde{s}_{r,j}$ before graph and FID-FIA refinement.

The FID-FIA refinement modifies only the existence branch. Let a_j denote the Fisher-information-distance accumulation from existence-weighted local target pairs, and define

$$q_{\mathrm{fid},j} = \rho_{\min} + (1-\rho_{\min}) \left(\frac{a_j}{\epsilon + \max_{u \in \mathcal{A}_{k,s}} a_u} \right)^{p_f}. \tag{44}$$

For positive $q_{\mathrm{fid},j}$, this contributes $\gamma_f \log q_{\mathrm{fid},j}$ to the existence utility and leaves the spatial utility unchanged. The main zero-floor setting is the boundary case in which a sensor with zero FID-FIA accumulation can receive zero instantaneous existence weight.

The structure-aware refinement then multiplies the spatial score by $\xi_{x,j}^{\gamma_x}$ and the existence score by $\xi_{r,j}^{\gamma_r}$.

Proposition C6. Let p_x and p_r be normalized priors satisfying

$$p_{x,j} \propto \xi_{x,j}^{\gamma_x/\tau_x}, \tag{45}$$

$$p_{r,j} \propto \xi_{r,j}^{\gamma_r/\tau_r}. \tag{46}$$

Then the branch-wise allocations are equivalently written as

$$w_x^\star = \arg\max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j \hat{u}_{x,j} - \tau_x D_{\mathrm{KL}}(w \,\|\, p_x) \right\}, \tag{47}$$

$$w_r^\star = \arg\max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j \hat{u}_{r,j} - \tau_r D_{\mathrm{KL}}(w \,\|\, p_r) \right\}. \tag{48}$$

Thus the weak structure-aware term is a KL regularization toward a graph-induced prior on the simplex.

Corollary C6a. With the FID-FIA existence refinement, the same program remains valid after replacing $\hat{u}_{r,j}$ by $\hat{u}^{\text{fid}}_{r,j} = \hat{u}_{r,j} + \gamma_f \log q_{\text{fid},j}$. Equivalently,

$$w^{\star}_{r,j} \propto \tilde{s}_{r,j} \xi^{\gamma_r}_{r,j} q^{\gamma_f}_{\text{fid},j}, \qquad w^{\star}_{x,j} \propto \tilde{s}_{x,j} \xi^{\gamma_x}_{x,j}. \tag{49}$$

The FID-FIA term is therefore an existence-branch log-utility, not a change to the spatial KLA barycenter.

Proposition C7. Define

$$U_x(w_x) = \sum_j w_{x,j} \hat{u}_{x,j} + \tau_x H(w_x), \tag{50}$$

$$U_r(w_r) = \sum_j w_{r,j} \hat{u}_{r,j} + \tau_r H(w_r), \tag{51}$$

with the FID-FIA substitution in U_r when Cardinality-critical mode is active. Under the aligned-label and projection assumptions above, the branch-decoupled adaptive weights solve

$$\max_{w_x, w_r \in \Delta_{k,s}} \left\{ U_x(w_x) + U_r(w_r) \right\}. \tag{52}$$

Together with Proposition C5, this gives a block-relaxed interpretation: the state-update block reproduces the aligned Bernoulli/LMB surrogate, and the weight-update block reproduces the branch-wise adaptive allocations.

Scalar-FID-FIA as a constrained special case. A single FID-FIA scalar weight for the whole fused posterior imposes $w_x = w_r = w_{\text{fid}}$ and lets the Fisher-separability utility influence both the Gaussian barycenter and the Bernoulli logit pool. The proposed refinement relaxes that constraint: w_x remains governed by the communication-aware spatial utility, while w_r receives the FID-FIA term. This explains why a scalar FID-FIA baseline can improve cardinality while perturbing localization; the branch-decoupled version reserves the information-geometric cue for existence/cardinality fusion.

C.5. Existence Confidence and Temporal Stabilization

The existence-confidence term uses $c(r) = |2r - 1|$, which is bounded in $[0, 1]$ and monotone in the magnitude of the Bernoulli natural parameter:

$$\left| \log \frac{r}{1-r} \right| = \log \frac{1 + c(r)}{1 - c(r)} = 2 \operatorname{artanh}(c(r)). \tag{53}$$

Thus existence confidence is a bounded proxy for the Bernoulli decisiveness appearing in (34).

The temporal stabilization step computes

$$w^{\text{ema}}_k = \alpha w_{k-1} + (1 - \alpha) \hat{w}_k. \tag{54}$$

Proposition C8. For any $\alpha \in [0, 1]$, w^{ema}_k is the minimizer of

$$\min_w \left\{ \alpha \| w - w_{k-1} \|_2^2 + (1 - \alpha) \| w - \hat{w}_k \|_2^2 \right\}. \tag{55}$$

The EMA step is therefore a quadratic compromise between the previous weights and the instantaneous weights. The subsequent minimum-weight enforcement remains a pragmatic feasibility projection rather than part of one single exact objective.

CRediT authorship contribution statement

Hao Lang: Conceptualization, Supervision, Project administration, Writing - review and editing. **Jinhao Chen**: Software, Validation, Formal analysis, Writing - original draft. **Tianyu Wo**: Supervision, Writing - review and editing.

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Data availability

No real-world datasets were used in this study. The simulation data supporting the findings are synthetic and can be regenerated from the MATLAB/Octave scripts and deterministic seeds used in the experiments. The code and generated result logs will be made available by the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT by OpenAI to improve the clarity, grammar, and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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